

UNIT IV

The Memory Organization

- ① Explain the basic memory hierarchy in Computer systems with a neat diagram.

"In the design of the Computer system, 'a processor', as well as a large amount of memory devices, has been used. However, the main problem is, these parts are expensive. So, the 'memory organization' of the system can be done by memory hierarchy."

Memory hierarchy:- The memory in a Computer can be divided into five hierarchies based on the speed as well as use. The processor can move from one level to another based on its requirements. The five hierarchies in the memory are registers, cache, main memory, magnetic discs, and magnetic tapes. The first three are volatile memories which mean when there is no power, and then automatically they lose their stored data. Whereas the last two hierarchies are not volatile which means they store the data permanently.

A memory element is the set of 'storage devices' which stores the binary data in the types of bits. In general, 'the storage memory' can be classified into two categories such as volatile as well as non-volatile.

* Memory Hierarchy in Computer Architecture:-

The memory hierarchy in Computer system mainly includes different storage devices. Most of the computers were inbuilt with extra storage to run more powerfully beyond the main memory capacity.

The following memory hierarchy diagram is a hierarchical pyramid for Computer memory. The designing of the memory hierarchy is divided into two types such as

- Primary (Internal) memory &
- Secondary (External) memory.

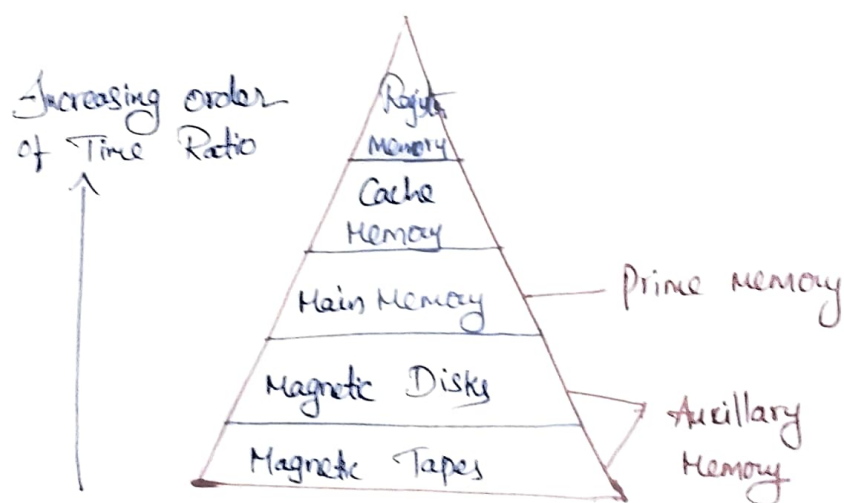


Fig: Memory Hierarchy

- Primary Memory:- The primary memory is also known as internal memory, and this is accessible by the processor straightly. This memory includes main, cache, as well as CPU registers.

→ Secondary memory:-

It is also known as external memory, and this is accessible by the processor through an input/output module. This memory includes an optical disk, magnetic disk, and magnetic tape.

Characteristics of Memory hierarchy:- Includes following

1. Performance:- Previously, the designing of a Computer system was done without memory hierarchy, and the speed gap among the main memory as well as the CPU registers enhances, because of the ^{huge} disparity in access time, which will cause the lower performance of the system. So enhancement was mandatory. The enhancement of this was designed in the memory hierarchy model due to the system's performance increase.
2. Ability:- The ability of the memory hierarchy is the total amount of data the memory can store. Because whenever we shift from top to bottom inside the memory hierarchy, then the capacity will increase.
3. Access Time:- The access time in the memory hierarchy is the interval of the time among the data availability as well as request to read or write. Because whenever we shift from top to bottom inside the memory hierarchy, then the access time will increase.

4. Cost per bit:- When we shift from bottom to top inside the memory hierarchy, then the cost for each bit will increase which means an internal memory is expensive compared with external memory.

Memory Hierarchy Design:-

The memory hierarchy in computers mainly includes the following.

Registers: Usually, the register is a static RAM or SRAM in the processor of the computer which is used for holding the ~~word~~ data word which is typically 64 or 128 bits. The program counter register is the most important as well as found in all the processors. Most of the processors use a status word register as well as an accumulator. A status word register is used for decision making, and the accumulator is used to store the data like mathematical operation. Usually, computers like 'Complex instruction set computers' have so many registers for accessing main memory, and 'RISC-reduced instruction set' computers have more registers.

Cache memory:- Cache memory can also be found in the processor, however rarely it may be another IC (Integrated Circuit). ~~which is~~

Main Memory:- The main memory in the Computer is nothing but, the memory unit in the CPU that communicates directly. It is the main storage unit of the Computer. This memory is fast as well as large memory used for storing the data throughout the operations of the Computer. This memory is made up of RAM as well as ROM.

Magnetic Disks:- The magnetic disks in the Computer are circular plates fabricated of plastic otherwise metal by magnetized material. Frequently, two faces of the disk are utilized as well as many disks may be stacked on one spindle by read or write heads obtainable on every plane. All the disks in Computer turn jointly at high speed. The tracks in the Computer are nothing but, bits which are stored within the magnetized plane in spots next to concentric circles. They are usually separated into sections which are named as sectors.

Magnetic Tape:- This tape is a normal magnetic recording which is designed with a slender magnetizable covering on an extended, plastic film of the strip. This is mainly used to back up huge data. Whenever the Computer requires to access a strip, first it will move to access the data. Once the data is

allowed, then it will be unmounted. The access time of memory will be slower within magnetic strip as well as it will take a few minutes for accessing a strip.

Advantages of memory hierarchy:-

1. Memory distributing is simple and economical
2. Removes external destruction.
3. Data can be spread all over.
4. Permits demand paging & pre-paging.
5. Swapping will be more proficient.

② Explain the working principle of Static RAM (SRAM) with a neat diagram. Compare it with Dynamic RAM (DRAM).

Static Random Access Memory (SRAM) is a type of semiconductor memory that stores data using bistable latching circuits (flip-flops). Unlike dynamic RAM (DRAM), which needs periodic refreshing, SRAM retains data as long as power is supplied.

Basic principle:-

Each bit of data in SRAM is stored using a flip-flop made of transistors.

→ The flip-flop can be in one of two stable states —

representing logic '1' or '0'

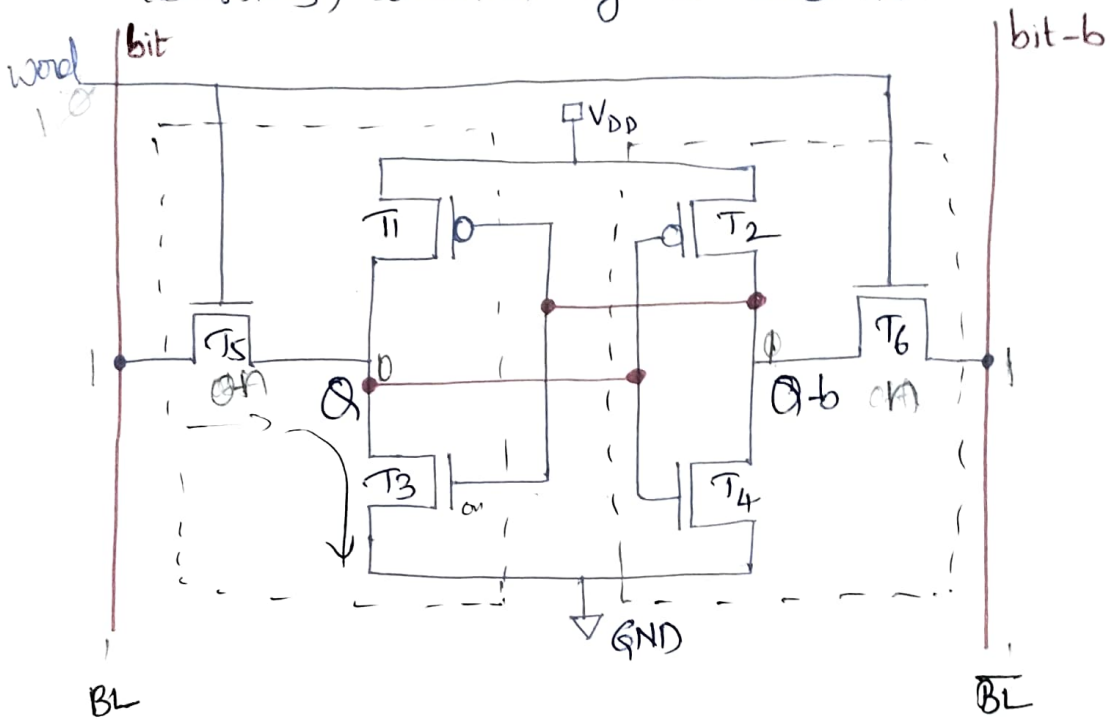
→ As long as power is on, the state remains unchanged — hence called static.

SRAM Cell Structure:-

A typical SRAM cell uses 6 transistors (6T).

→ 4 transistors ($T_1 - T_4$) from two cross-coupled inverters (2 flip-flop)

→ 2 transistors (T_5, T_6) act as access transistors (or pass transistors) controlled by the word line.



operation of SRAM:-

write operation:

- The desired data ('0' or '1') is placed on the bit-lines (BL or $\bar{BL}$). For example, if writing '1', BL might be driven high and $\bar{BL}$ driven low.
- The word-line (WL) is asserted (goes high), turning on the access transistors. This connects the bit-lines to the internal nodes of the latch.
- The driver on the bit-lines overpowers the existing latch state, forcing the internal nodes to the values ($Q=1, \bar{Q}=0$). The cross-coupled inverters then latch

This state.

[

1. Assume $Q=0$ and we wish to write a 1 into the cell.
2. bit = high, bit-b = low & word = low
3. Now word = high.]

Read Operation:-

- Bit lines are (BL & $\overline{BL}$) are pre-charged (e.g. to V_{DD}).
- The word-line is asserted, turning on the access transistor and connecting internal nodes to the bit-lines.
- If one internal node (say Q is high) & other (Q_b) is low, one of the bit lines will discharge relative to other.
- After read, the cell remains intact (no data lost) bcz latch structure maintains its state.

[

~~Assume~~ ^{pre} $Q=0$ and we wish to write a 1 into the cell.

1. Initially bit & bit-b = high, word = low
2. Assume $Q=0$, thus $Q_b=1$
3. Now word = high.]

Advantages:

1. Very fast access.
2. Does not require refresh cycles (unlike DRAM). The data remains as long as power is applied.
3. Simpler interface for high-speed caches and registers.

Disadvantages:-

- Because each bit uses more transistors (e.g 6T), the density (bits per data) is lower than DRAM.
- The cost per bit is higher.
- It is still volatile memory: If power is removed, data is lost.

Differences b/w SRAM and DRAM:-

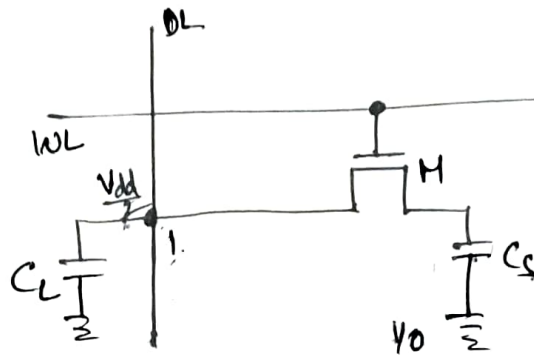
<u>Feature</u>	<u>SRAM</u>	<u>DRAM</u>
Cell Structure	6 transistors per bit	1 transistor + 1 Capacitor per bit
Refresh requirement	No periodic refresh needed.	Requires periodic refresh of stored charge.
Access speed	fast	Slow compared to SRAM
Density (bits per area)	low (takes more area)	high (more bits per chip).
Cost per bit	more expensive	Cheaper for large capacity
Power Consumption	low	high
Application	Cache memory, small fast memory close to CPU	Main system memory (RAM Controller..)

③ Describe the organization and working of DRAM cells.
Why does DRAM requires refreshing?

DRAM - Dynamic Random Access memory.

DL - Data Line
WL - Word Line

1T - 1C1R1



"here DL act as o/p"

Read Operation

- 1) $C_L = \frac{V_{dd}}{2}$, we need to make $C_L = \frac{V_{dd}}{2}$
- 2) $W_L = 1$, $M = ON$, M is ON , then it completely depend on C_S
if $C_S = 1V$, then , then C_S voltage is send to node 1, so
voltage node 1 $V > \frac{V_{dd}}{2} = DL$

Sense amplifier

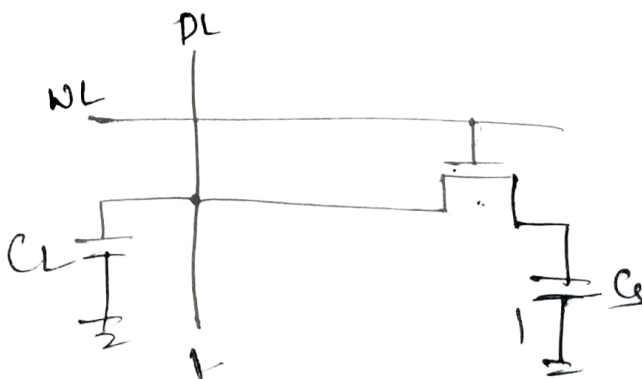
↓
o/p = 1.

★

$V < \frac{V_{dd}}{2}$, o/p = 0

After this refresh the cell.

write Operation



"here DL act as i/p"

If we place '1' at DL, the C_S is 1 else C_S is '0'.

1 → charge
0 → discharge.

1) $DL = 1$

2) $WL = 1$

3) $DL = 1 \rightarrow C_s = V_{dd}$ (charging)

$= 0 \rightarrow C_s = 0$ (discharging)

Refreshing in DRAM:—

- process of reading the information of memory & immediately re-writing the same without any modification.

"DRAM requires refreshing because it stores data as electrical charge in capacitors, and these charges naturally leak away with time. Refresh cycles store the charge to preserve data integrity."

Q.2 compare RAM, ROM in terms of speed, volatility, and use cases?

	RAM (Random Access Memory)	ROM (Read Only Memory)
1) Speed →	Very fast, data can be read & write very fast.	Slower compared to RAM, mainly used for reading data.
2) Volatility	Volatile - data is lost when power is turned off.	Non-volatile - data is retained even without power.
3) Use cases	used as temporary working memory for CPU during program execution.	used for storage (Permanent)
4) Data access	Read and write are possible.	Generally read-only (cannot be easily modified)
5) Examples	SRAM, DRAM	PROM, EPROM, EEPROM

Q.3 Explain different cache mapping techniques

(i) Direct mapping (ii) Associative mapping

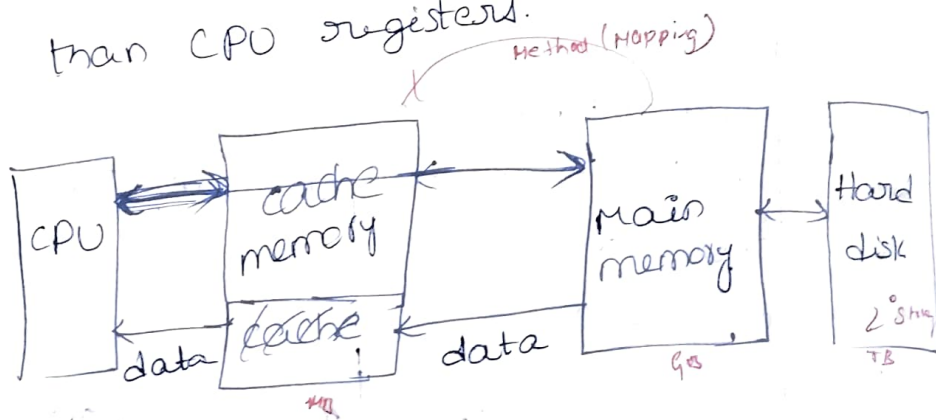
(iii) Set-associative mapping.

A:- cache memory acts as a temporary storage area where computer processor can easily retrieve data and it can act as

butter between RAM and CPU.

Purpose of cache memory:-

- 1) It can be used to speed up and synchronize with high-speed CPU.
- 2) It saves frequently requested data and instructions so that they can be used to CPU immediately when needed.
- 3) It is more expensive than main memory or disk memory but less expensive than CPU registers.



cache hit ratio & calculation.

cache hit ratio is the measure of how efficiently the cache memory is being used.

cache hits : when CPU finds the required data in cache.

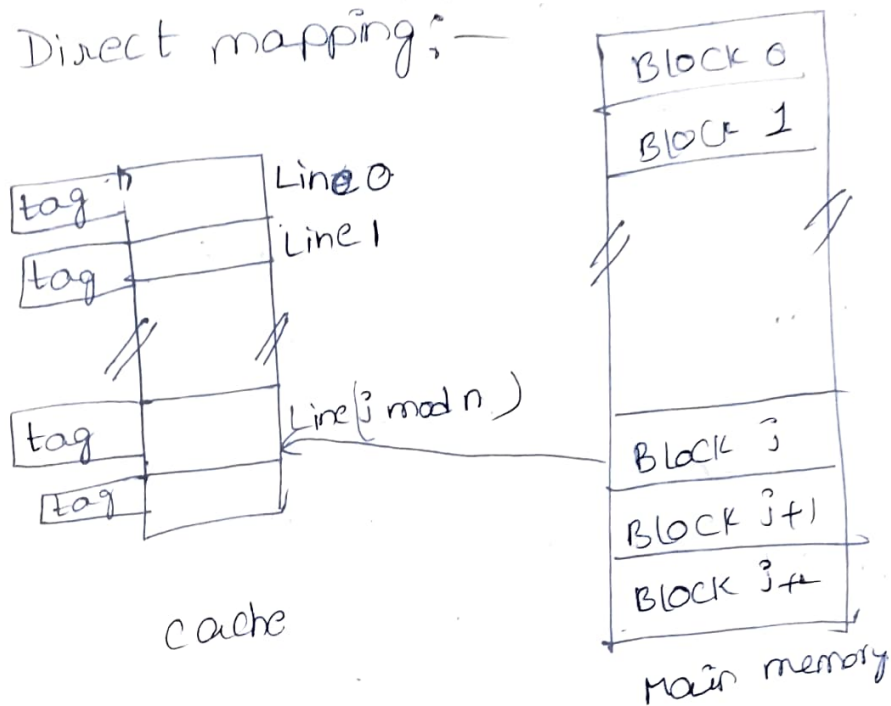
cache miss : when CPU doesn't find the required data in cache

$$\text{cache hit ratio} = \frac{\text{hits}}{\text{hits} + \text{misses}} \times 100$$

Mapping:- is the process of

transferring required word (or) data from main memory to cache.

(1) Direct mapping:-



→ It is the simplest cache memory technique where each main memory block is mapped to one specific, predetermined cache line. To find block, the main memory address is divided into a tag, index and offset. The index part of the address determines the exact cache line where the block must reside; if this line is already occupied by another block, the old one is replaced.

How it works:- A memory address is split into three parts. Tag, index and word offset.

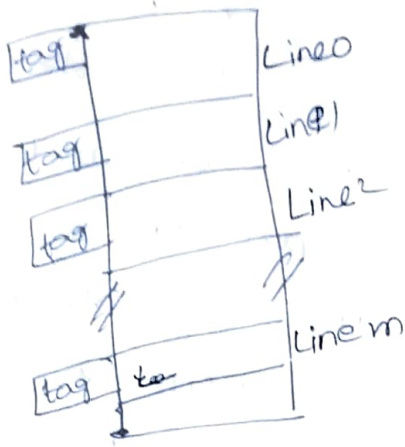
Tag: The most significant bits, used to verify the correct block is in the cache line.

Index: The middle bits, which specify the exact cache line number where the block is stored.

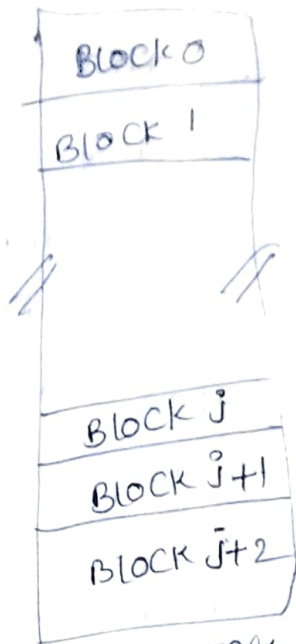
word offset: The least significant bits, which locate the specific word within the block.

Mapping process: when a memory block is accessed, its address is sent to the cache. The index bits are used to select a specific cache line. If the tags match and the line is valid, a "hit" occurs and the requested word is retrieved. If the tags do not match, a "miss" occurs. The requested block is then fetched from main memory and replaces the old block.

2) **Associative mapping:** It allows any main memory block to be placed in any available cache line, offering high flexibility and a better hit ratio.



cache



main memory

This is achieved by using a tag to identify the memory block, which is compared against all other tags in the cache simultaneously.

How it works:-

Any block of main memory can be mapped to any line in the cache. There is no fixed location for a memory block. The main memory address is split into a tag and a word offset. There is no "index" field, as the block can be anywhere in the cache. While this technique is fast, it is also expensive to implement due to the complex hardware required for parallel tag comparison.

3) **Set associative mapping**:- This is a hybrid approach that combines the simplicity of direct mapping with the flexibility of fully associative mapping to reduce conflict misses. The cache memory divided

into sets, allowing a memory block to be placed in any line within a specific set.

The cache is organized into sets, and the main memory address is split into a "set number" a "tag" and a "byte offset". when searching for data, the system first finds the correct set using the set number, then checks all lines within that set for a matching tag.

Q. With diagrams, explain the working of EPROM and EEPROM.

ROM: ROM stands for Read-only Memory. It is a type of memory that retains its contents even when the power is off. For this reason, ROM is also called non-volatile memory.

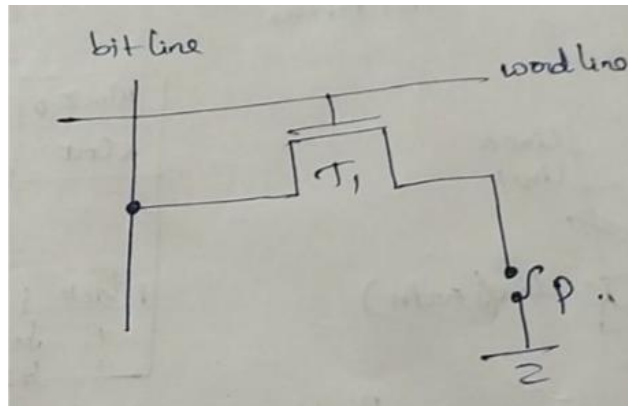
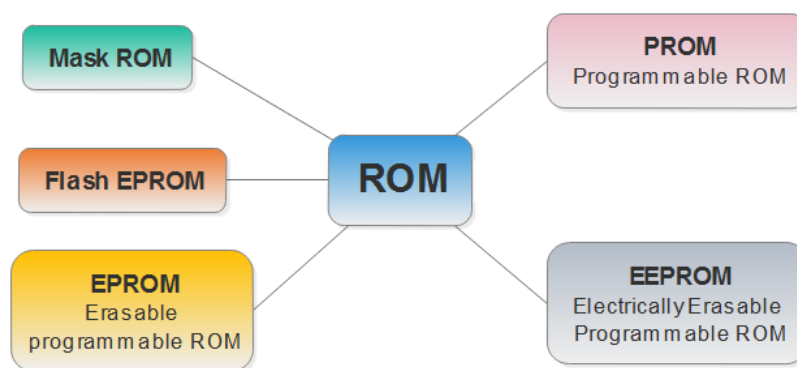


Fig: ROM

Although all ROM basically serves the same purpose, there are a few different types commonly in use today.

The different types of ROM used in the industry are:

1. **PROM (Programmable ROM)**
2. **EPROM (Erasable Programmable ROM)**
3. **EEPROM (electrically erasable programmable ROM)**
4. **Flash EPROM**
5. **Mask ROM**



1. PROM (programmable ROM)

PROM refers to the kind of ROM that the user can burn information into. In other words, PROM is a **user-programmable memory**.

For every bit of the PROM, there exists a fuse. PROM is programmed by blowing the fuses. If the information burned into PROM is wrong, that PROM must be discarded since its internal fuses are blown permanently. For this reason, PROM is also referred to as OTP (**One Time Programmable**).

Programming ROM also called burning ROM, requires special equipment called a ROM burner or ROM programmer.

2.EPROM (erasable programmable ROM)

EPROM was invented to **allow making changes** in the contents of PROM after it is burned.

In EPROM, one can program the memory chip and erase it thousands of times. This is especially necessary during the development of the prototype of a **microprocessor-based project**.

A widely used EPROM is called **UV-EPROM**, where UV stands for ultraviolet. The only problem with UV-EPROM is that erasing its contents can take up to 20 minutes.

All UV-EPROM chips have a window through which the programmer can shine ultraviolet (UV) radiation to erase the chip's contents. For this reason, EPROM is also referred to as **UV-erasable EPROM** or simply UV-EPROM.

Programming a UV-EPROM

To program a UV-EPROM chip, the following steps must be taken:

1. Its contents must be erased. To erase a chip, remove it from its socket on the system board and place it in EPROM erasure equipment to expose it to UV radiation for 5-20 minutes.
2. Program the chip. To program a UV-EPROM chip, place it in the ROM burner (programmer). To burn code or data into EPROM, the ROM burner uses 12.5 volts or higher, depending on the EPROM type. This voltage is referred to as V_{pp} in the UV-EPROM datasheet.
3. Place the chip back into its socket on the system board.

As can be seen from the above steps, not only is there an EPROM programmer (burner), but there is also separate EPROM erasure equipment.

“The main problem, and indeed the major disadvantage of UV-EPROM, is that it cannot be erased and programmed while it is in the system board. To provide a solution to this problem, EEPROM was invented.”

3.EEPROM (electrically erasable programmable ROM)

EEPROM has several advantages over EPROM, such as the fact that its method of erasure is electrical and therefore instant as opposed to the 20-minute erasure time required for UV-EPROM.

In addition, in EEPROM one **can select which byte to be erased**, in contrast to UV-EPROM, in which the entire contents of ROM are erased.

“However, the main advantage of EEPROM is that one can program and erase its contents while it is still in the system board. It does not require the physical removal of the memory chip from its socket. In other words, unlike UV-EPROM, EEPROM does not require an external erasure and programming device.”

4.Flash Memory EPROM

Since the early 1990s, Flash EPROM has become a popular user-programmable memory chip. and for good reasons.

1. First, the erasure of the entire contents takes less than a second, or one might say in a flash, hence its name, Flash memory.
2. In addition, the erasure method is electrical, and for this reason, it is sometimes referred to as Flash EEPROM. To avoid confusion, it is commonly called Flash memory.

“The major difference between EEPROM and Flash memory is that when Flash memory’s contents are erased, the entire device is erased, in contrast to EEPROM, where one can erase the desired byte.”

5.Mask ROM

Mask ROM refers to a type of ROM in which the contents are programmed by the IC manufacturer. In other words, it is not a user-programmable ROM.

The term mask is used in [IC fabrication](#). Since the process is costly, mask ROM is used when the needed volume is high (hundreds of thousands) and it is absolutely certain that the contents will not change.

“It is common practice to use UV-EPROM or Flash for the development phase of a project, and only after the code/data have been finalized is the mask version of the product ordered.”

Q. Discuss the role of memory interleaving in improving performance.

Memory Interleaving is a technique that spreads memory addresses evenly across multiple memory banks, allowing parallel access to memory and thus reducing latency. Instead of accessing memory sequentially from one module, interleaving allows the system to access different modules simultaneously.

What Is Memory Interleaving?

- Memory is divided into **banks** or **modules**. When interleaving is applied, memory addresses are mapped across these banks in a round-robin fashion.

For example, if you have 4 memory banks and data stored at sequential memory addresses, interleaving would distribute them like this

Address	Memory Bank
-----	-----
0	0
1	1
2	2
3	3
4	0
5	1
..	..

By spreading memory accesses, the CPU can fetch multiple words in parallel, significantly reducing access times.

Types of Memory Interleaving

1. Low-Order Interleaving

- Uses the **least significant bits** of the address to select the bank.
- Ideal for systems that perform **sequential memory access**.
- Ensures that consecutive addresses go to different banks, enabling parallel access.

2. High-Order Interleaving

- Uses the most significant bits of the address to select the bank.
- Often used in multi-processor systems where each processor accesses a distinct region of memory.
- Reduces contention when processors are accessing different memory ranges.

3. Adaptive interleaving

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- Advanced systems adjust their interleaving strategy dynamically based on observed access patterns for optimal performance.

Advantages of Memory Interleaving

- Reduced effective memory access time.
- Better system throughput and CPU utilization.
- Increased memory bandwidth.
- Efficient support for fast processors and pipelined architectures.

Limitations

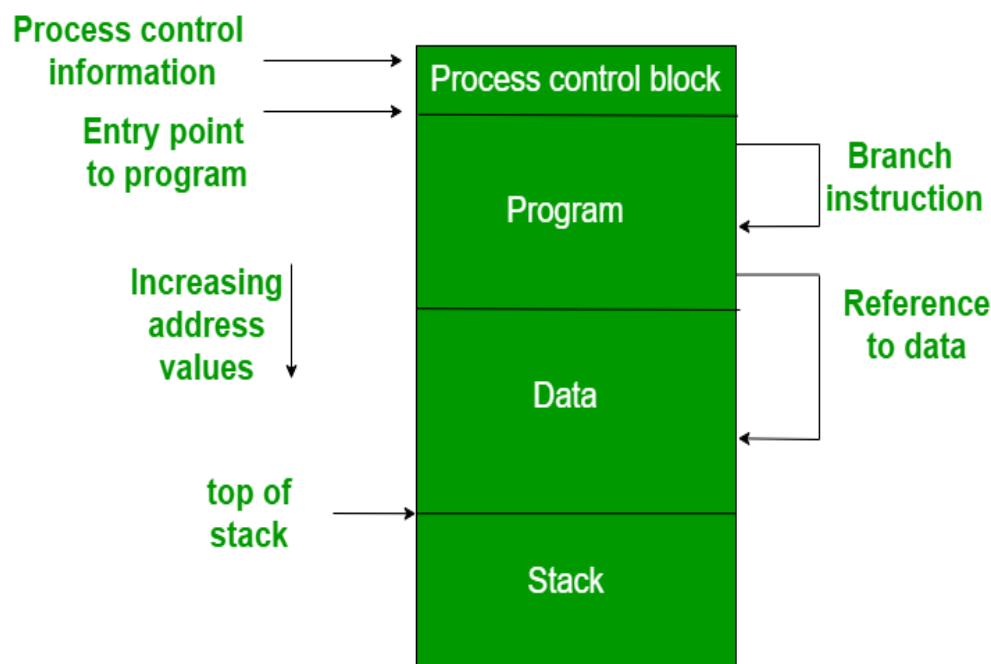
- Increased hardware complexity.
- Requires careful address mapping and synchronization.
- Performance benefits depend on access patterns; random access patterns may not fully utilize interleaving.

Q. Explain the requirements of memory management in modern operating systems.

Memory management keeps track of the status of each memory location, whether it is allocated or free. It allocates the memory dynamically to the programs at their request and frees it for reuse when it is no longer needed. Memory management meant to satisfy some requirements that we should keep in mind. These Requirements of memory management are:

Relocation - The available memory is generally shared among a number of processes in a multiprogramming system, so it is not possible to know in advance which other programs will be resident in main memory at the time of execution of this program. Swapping the active processes in and out of the main memory enables the operating system to have a larger pool of ready-to-execute process.

When a program gets swapped out to a disk memory, then it is not always possible that when it is swapped back into main memory then it occupies the previous memory location, since the location may still be occupied by another process. We may need to **relocate** the process to a different area of memory. Thus there is a possibility that program may be moved in main memory due to swapping.



The figure depicts a process image. The process image is occupying a continuous region of main memory. The operating system will need to know many things including the location of process control information, the execution stack, and the code entry. Within a program, there are memory references in various instructions and these are called logical addresses. After loading of the program into main memory, the processor and the operating system must be able to translate logical addresses into physical addresses. Branch instructions contain the address of the next instruction to be executed. Data reference instructions contain the address of byte or word of data referenced.

Protection - There is always a danger when we have multiple programs at the same time as one program may write to the address space of another program. So every process must be protected against unwanted interference when other process tries to write in a process whether accidental or incidental. Between relocation and protection requirement a trade-off occurs as the satisfaction of relocation requirement increases the difficulty of satisfying the protection requirement.

Prediction of the location of a program in main memory is not possible, that's why it is

impossible to check the absolute address at compile time to assure protection. Most of the programming language allows the dynamic calculation of address at run time. The memory protection requirement must be satisfied by the processor rather than the operating system because the operating system can hardly control a process when it occupies the processor. Thus it is possible to check the validity of memory references.

Sharing - A protection mechanism must have to allow several processes to access the same portion of main memory. Allowing each processes access to the same copy of the program rather than have their own separate copy has an advantage. For example, multiple processes may use the same system file and it is natural to load one copy of the file in main memory and let it shared by those processes. It is the task of Memory management to allow controlled access to the shared areas of memory without compromising the protection. Mechanisms are used to support relocation supported sharing capabilities.

1. Logical organization - Main memory is organized as linear or it can be a one-dimensional address space which consists of a sequence of bytes or words. Most of the programs can be organized into modules, some of those are unmodifiable (read-only, execute only) and some of those contain data that can be modified. To effectively deal with a user program, the operating system and computer hardware must support a basic module to provide the required protection and sharing.

It has the following advantages:

- Modules are written and compiled independently and all the references from one module to another module are resolved by the system at run time.
- Different modules are provided with different degrees of protection.
- There are mechanisms by which modules can be shared among processes. Sharing can be provided on a module level that lets the user specify the sharing that is desired.

2. Physical organization - The structure of computer memory has two levels referred to as main memory and secondary memory. Main memory is relatively very fast and costly as compared to the secondary memory. Main memory is volatile. Thus secondary memory is provided for storage of data on a long-term basis while the main memory holds currently used programs. The major system concern between main memory and secondary memory is the flow of information and it is impractical for programmers to understand this for two reasons:

- The programmer may engage in a practice known as overlaying when the main memory available for a program and its data may be insufficient. It allows different modules to be assigned to the same region of memory. One disadvantage is that it is time-consuming for the programmer.

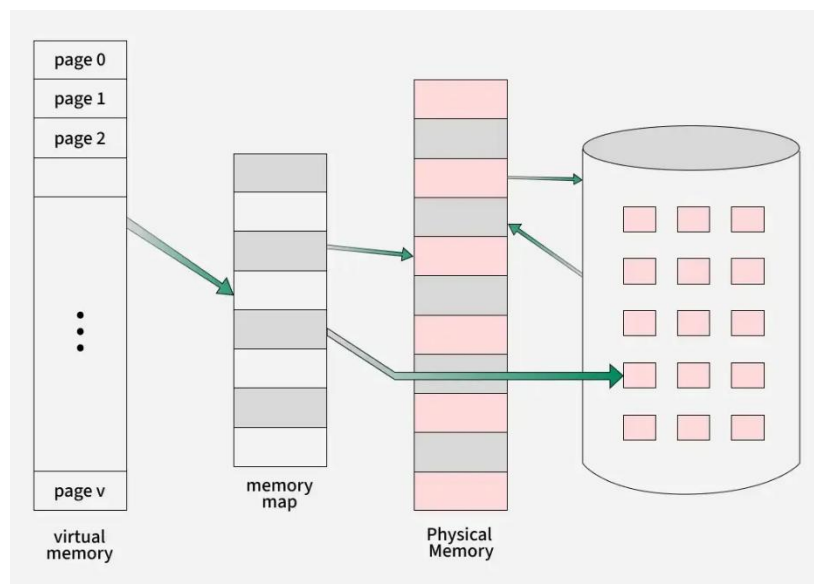
- In a multiprogramming environment, the programmer does not know how much space will be available at the time of coding and where that space will be located inside the memory.

Q. Explain the concept of virtual memory with a neat diagram.

Virtual memory is a memory management technique used by operating systems to give the appearance of a large, continuous block of memory to applications, even if the physical memory (RAM) is limited and not necessarily allocated in contiguous manner. The main idea is to divide the process in pages, use disk space to move out the pages if space in main memory is required and bring back the pages when needed.

Objectives of Virtual Memory

- A program doesn't need to be fully loaded in memory to run. Only the needed parts are loaded.
- Programs can be bigger than the physical memory available in the system.
- Virtual memory creates the illusion of a large memory, even if the actual memory (RAM) is small.
- It uses both RAM and disk storage to manage memory, loading only parts of programs into RAM as needed.
- This allows the system to run more programs at once and manage memory more efficiently.



Working:

- Virtual memory uses both hardware and software to manage memory.
- When a program runs, it uses virtual addresses (not real memory locations).

- The computer system converts these virtual addresses into physical addresses (actual locations in RAM) while the program runs.

Types of Virtual Memory

In a computer, virtual memory is managed by the Memory Management Unit (MMU), which is often built into the CPU. The CPU generates virtual addresses that the MMU translates into physical addresses. There are two main types of virtual memory:

1. Paging
2. Segmentation

1. Paging

Paging divides memory into small fixed-size blocks called pages. When the computer runs out of RAM, pages that aren't currently in use are moved to the hard drive, into an area called a swap file. Here,

- The swap file acts as an extension of RAM.
- When a page is needed again, it is swapped back into RAM, a process known as page swapping.
- This ensures that the operating system (OS) and applications have enough memory to run.

2. Segmentation

Segmentation divides virtual memory into segments of different sizes. Segments that aren't currently needed can be moved to the hard drive. Here,

- The system uses a segment table to keep track of each segment's status, including whether it's in memory, if it's been modified and its physical address.
- Segments are mapped into a process's address space only when needed.

Applications of Virtual memory

Virtual memory has the following important characteristics that increase the capabilities of the computer system.

- Increased Effective Memory
- Memory Isolation
- Efficient Memory Management
- Simplified Program Development

Benefits of Using Virtual Memory

- Supports Multiprogramming & Larger Programs
- Maximizes Application Capacity
- Eliminates Physical Memory

- Boosts Security & Isolation
- Improves CPU & System Performance
- Enhances Memory Management Efficiency

Limitations of Virtual Memory

- Slower Performance
- Risk of Data Loss
- More Complex System

Q. Explain the different types of optical storage media (CD, DVD, Blu-Ray) with their characteristics.

As compared to magnetic memory, optical memory is a relatively new secondary storage medium. It has proved to be a promising random medium for high-capacity secondary storage, as it can store large amounts of text or data in limited space. An optical-disk storage system consists of a rotating disk, which is coated with a thin metal or any other material that is highly reflective. Laser beam technology is used for recording/reading data on a disk. Due to this, optical disks are also known as laser disks or optical laser disks.

The optical memory can be classified into three categories:

- CD (Compact Disk)
- DVD (Digital Video Disk)
- Blu-Ray Disk

Compact Disk

CD has a diameter of 12 cm. The track on the CD is spiral-shaped, with around 20,000 windings. We've already explored the notion of information storage within the substrate. It is affordable, portable, and has a storage capacity of approximately 700 Mb. A compact disk operates in such a way that when the disc is inserted in the disk tray of a CD player, the track of the disc is scanned by a low-intensity laser beam. By rotating the disc between 200 and 500 revolutions per minute, a consistent scanning pace is maintained.

It is classified into three ways.

- CD-ROM: Compact Disk Read Only Memory, on which the manufacturer has pre-stored data.
- CD-R: Compact Disk-R is recordable, the data within this is written only one time by the user but it is not rewritable.
- CD-RW: Compact Disk-RW is rewritable, the data can be modified multiple times i.e., this can be read, erased, and written again and again as well.

Digital Video Disk

DVD is also known as digital versatile disk. This is a different kind of optical disk that holds data and multimedia content. It is the second form of optical disc to be introduced, following the compact disk. Despite its appearance, it can hold 15 times the amount of data that a CD can. A DVD provides approximately 4.7 GB to 17 GB of storage space.

DVD is also classified into mainly three parts.

- DVD-ROM: DVD-Read Only Memory permits the manufacturer to store data/information within it at the time of disk manufacturing and does not allow the disk to be rewritten.
- DVD-R: DVD-Recordable data is available to the user in blank format, and the user can only store data in it once.
- DVD-RW: DVD-Rewritable provides users with the ability to read and write data many times.

Blu-ray Disk

It is regarded as the third generation of compact disk technology, following the launch of CDs and DVDs. It is the most recent optical storage medium for storing high-definition (HD) audio and video. It contains about 128 GB of storage space.

Unlike CDs and DVDs, which employ a red laser beam for read/write operations, blu-ray disks use a blue-violet laser with a wavelength of 405 nm. The tracks on the disk can be wound more closely due to the shorter wavelength of the laser than the high wavelength beam utilized by CDs and DVDs. As a result, it has a large storage capacity and can hold more data than CDs and DVDs.

Applications of Optical Storage Systems

Optical storage devices are commonly employed in data storage media.

- In Computers
- In Audio Players
- In Digital Cameras

Advantages of Optical Storage Systems

- The cost-per-bit of storage for optical disks is very low.
- The use of a single spiral track optical disks an ideal storage medium for reading large blocks of sequential data such as music.
- Due to their compact size and light weight, optical disks are easy to handle, store, and port from one place to another.

Disadvantages of Optical Storage Systems

- It is read-only (permanent) storage medium. Data once recorded, cannot be erased and hence, the optical disks cannot be reused.

- The data access speed for optical disks is slower than magnetic disks.
- It requires a more complicated drive mechanism than magnetic disks.